MPCA – Mini Project – 4th Sem B - Section	

MPCA Project

TITLE OF THE PROJECT: Digital Notice board with Scrolling LED display

Team Members:

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Components Used

- **Arduino Uno**
- **Breadboard**
- **Ultrasonic Sensor (HC-SR04)**
- **Temperature Sensor (DHT11)**
- **16x2 LCD with I2C module**
- **Jumper wires**
- **USB cable (for power & programming)**

Explanation about the project :

This project is designed to measure distance and temperature simultaneously using an Arduino-based system and display the results on an LCD.

The ultrasonic sensor (HC-SR04) is used to measure the distance of an object from the sensor by sending ultrasonic waves and calculating the time taken for the echo to return. The distance is computed using the speed of sound.

The DHT11 temperature sensor is used to measure the surrounding temperature. It provides digital output, which is read by the Arduino.

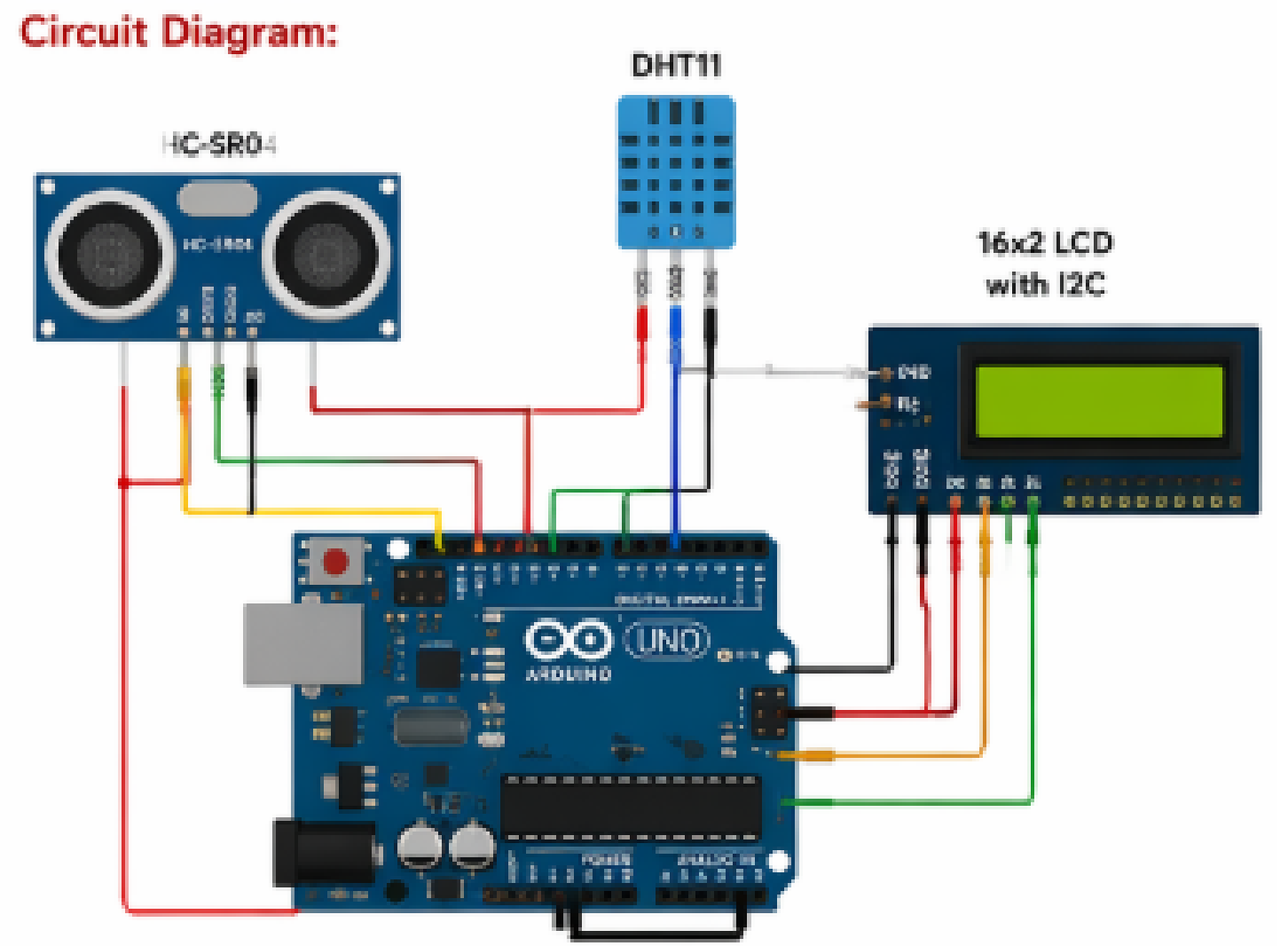
A 16x2 LCD with I2C interface is used to display the measured values. The I2C module reduces the number of connections by using only two communication lines (SDA and SCL).

The Arduino processes both sensor inputs and displays:

- Temperature in °C
- Distance in cm
- Status messages like *"Object is near"* or *"Area Clear"*

Additionally, the display uses a scrolling mechanism to show longer messages clearly on the LCD.

Circuit Diagram:



Pseudocode:

START

Initialize LCD

Initialize DHT sensor

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Set TRIG as OUTPUT

Set ECHO as INPUT



LOOP forever:

  Read temperature from DHT sensor

  IF temperature is valid:

    Store temperature

    IF temperature >= 34:

      Add HIGH warning

  ELSE:

    Show "Temp Error"

  Trigger ultrasonic sensor

  Measure echo time

  Calculate distance

  IF distance < 10:

    Display "Object is near"

  ELSE IF distance > 25:

    Display "Area Clear"

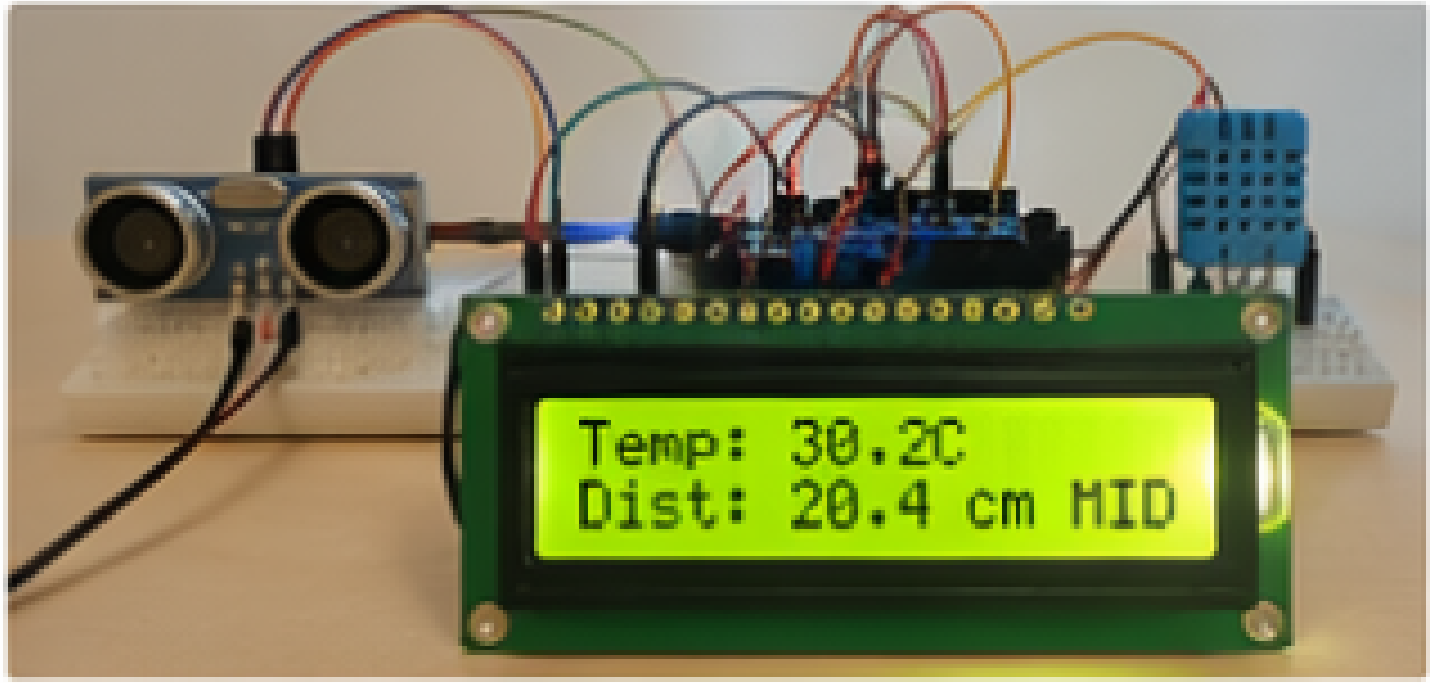
  ELSE:

    Display "MID distance"

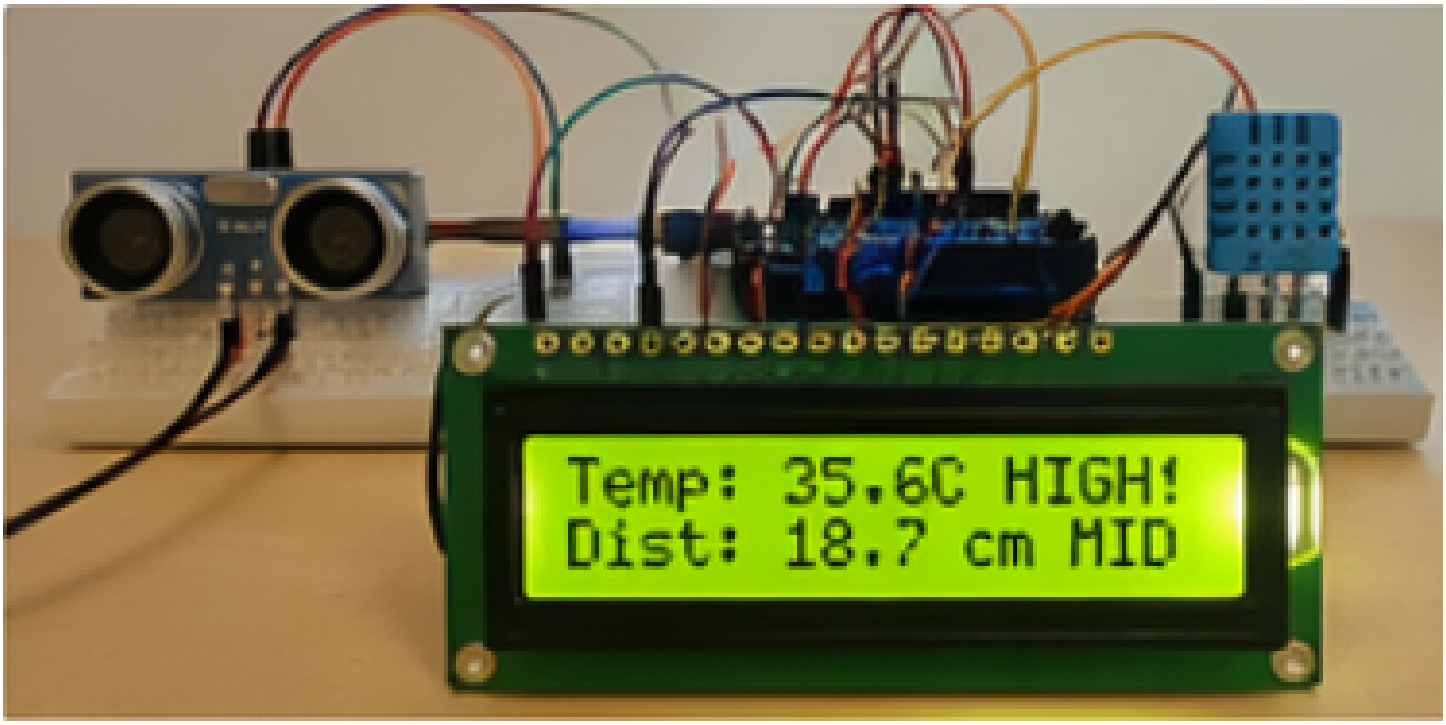
  Add spaces for scrolling
  Display both lines on LCD with scrolling effect
  Wait for 2 seconds
END LOOP
```

Output Pictures with explanations:

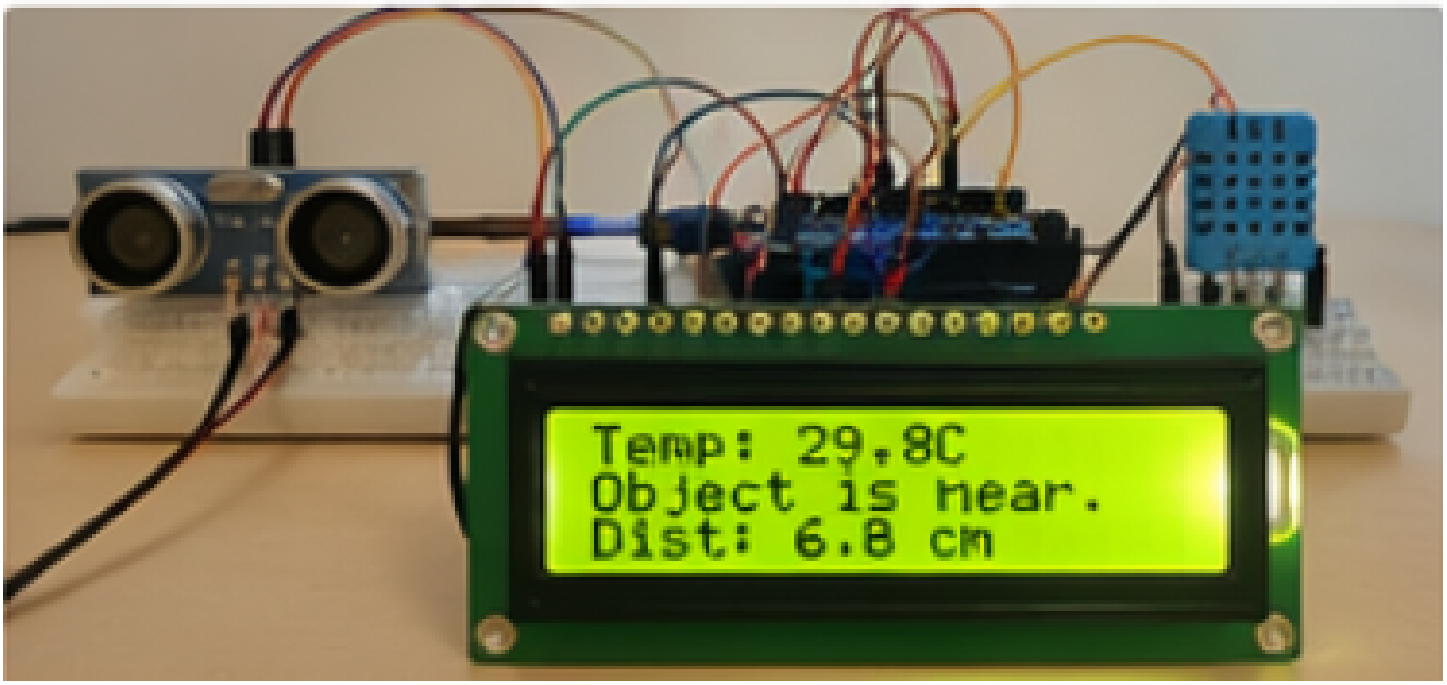
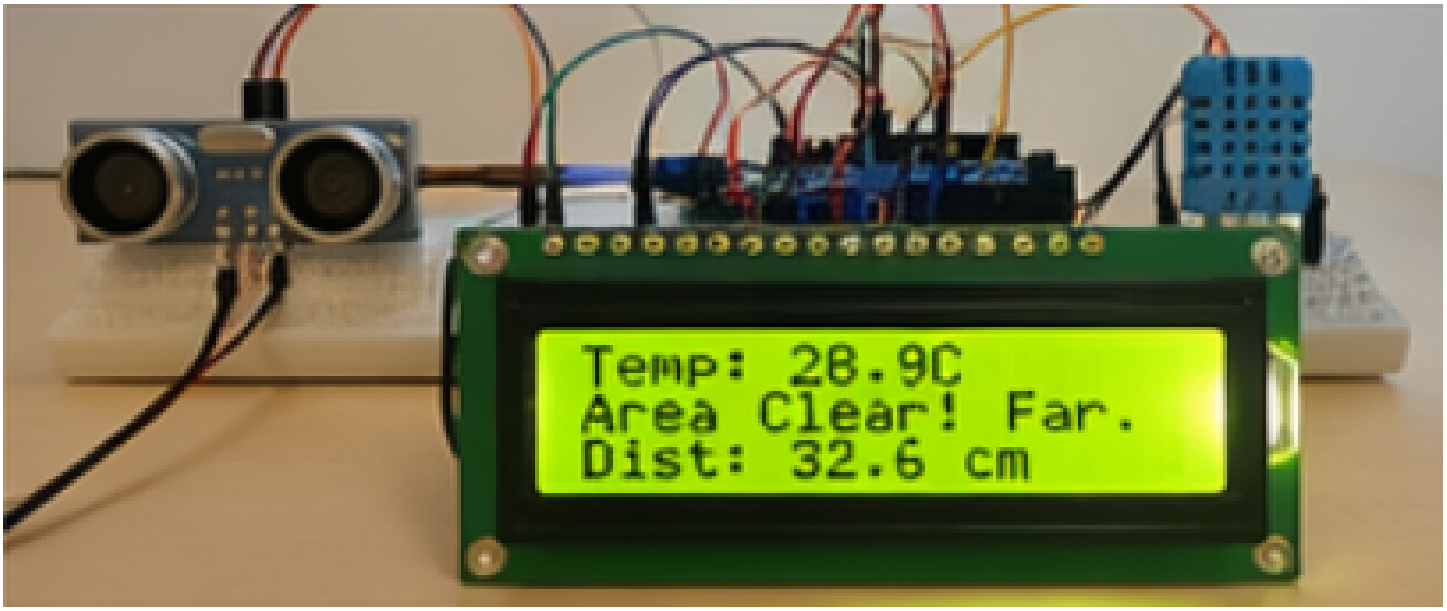
case1:-Normal case



case2:- High Temparature condition



case3:- Object near



case4:-object far